Weather Disruption Assessment SOP

SYN-SOP-02 | Version 1.0 | Status: Synthetic MVP guidance

Synthetic operational knowledge created for the Option C Logistics project demonstration. This document is not a real company policy and must not be used for live operational decisions.

Purpose	Guide evidence-based use of destination weather during shipment disruption investigations.
Applies to	Supply-chain disruption agent demonstration, operations review, and RAG retrieval testing.
Decision authority	Human operations reviewer. The AI system provides evidence and recommendations only.
Evidence rule	Use only observed tool evidence, model output, and retrieved guidance. Do not invent operational facts.

1. Weather evidence standard

- Geocode destination city, state, and country rather than assuming customer coordinates represent the destination.
- Record the geocoded location, observation window, weather source, and any route-level limitation.
- Use observed or forecast data appropriate to the shipment dates. Label historical observations clearly.
- Do not describe weather as severe unless the data and severity logic support that conclusion.

2. Severity interpretation

Low	Light precipitation, modest wind, or benign conditions. Continue monitoring; do not reroute solely on this evidence.
Moderate	Material precipitation, wind, or repeated adverse conditions. Review likely operational impact and timing.
High	Severe storms, extreme precipitation, dangerous wind, flooding, or official disruption signals. Escalate and assess alternatives.

3. Investigation steps

1. Confirm the shipment destination and scheduled delivery window.
2. Geocode the written destination and record the returned location.
3. Retrieve weather data for an appropriate time window.
4. Summarize precipitation, wind, observed condition, and severity.
5. Compare the severity with the model risk and historical delay context.

6 Recommend monitoring or transport review proportionate to the evidence.

4. Rerouting rule

Do not recommend immediate rerouting solely because weather data exists. Rerouting requires material weather severity or corroborating transport, inventory, supplier, or service evidence.

5. Required limitations

- Destination weather may not represent the full shipment route.
- Geocoding may be approximate where place names are ambiguous.
- Historical weather cannot confirm a live operational disruption.
- Weather evidence must not be presented as a supplier or inventory finding.